



Thesis Defense (30/06/2026)

Performance Optimization of the AI4Land Training Pipeline on MareNostrum 5

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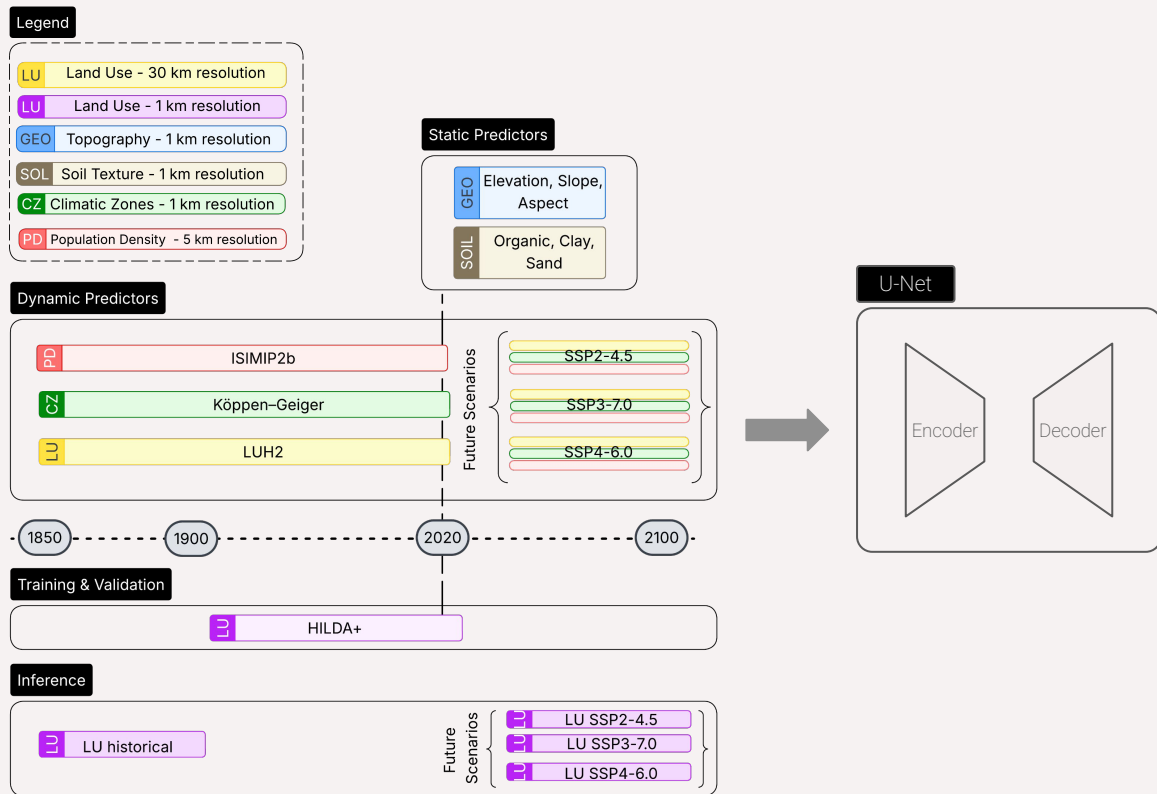
3: UAB

Introduction: AI4Land

Built on **CONCERTO**, TerraDT
Reused by **ELLIOT**

Resolution gap:

- LUH2 30 km; past, present and future
- HILDA+ 1 km, present



Objectives

MAIN

Achieve a faster training time

Secondary

Faster training,
same accuracy

Understand each
speedup

Reinvest the
savings

Reusable
guidelines

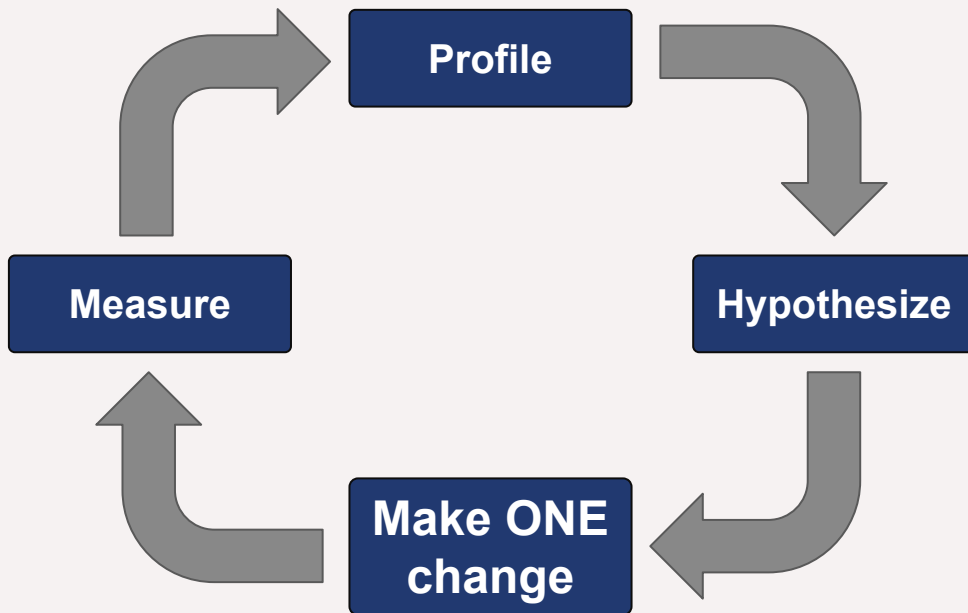
Methodology

Experiments

- Run on MareNostrum 5
- Shared machine
- Run N times
- Review all results

Tools

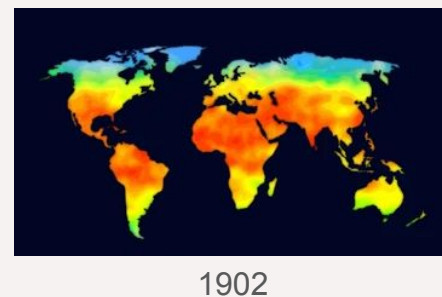
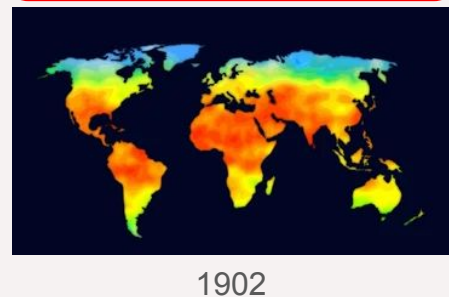
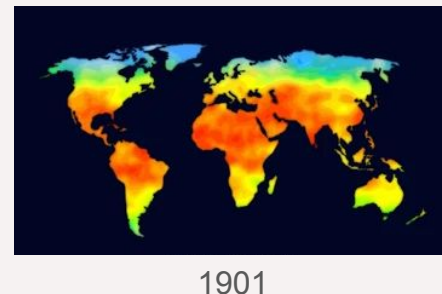
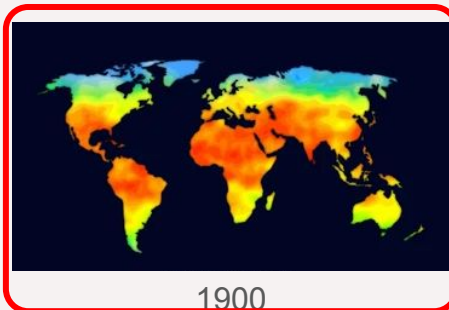
- NSys + NVTX
- Xarray
- PyTorch
- Zarr



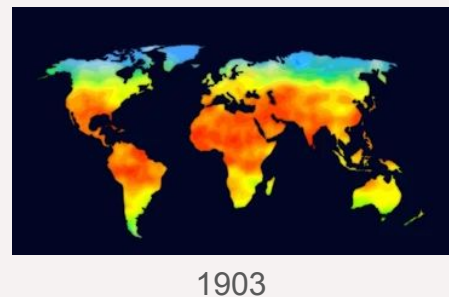
Zarr Stores

- Analysis-Ready
Cloud-Optimized (**ARCO**)
- Chunks data
- Stores > arrays > chunks
- Each store is a python object

Variable X

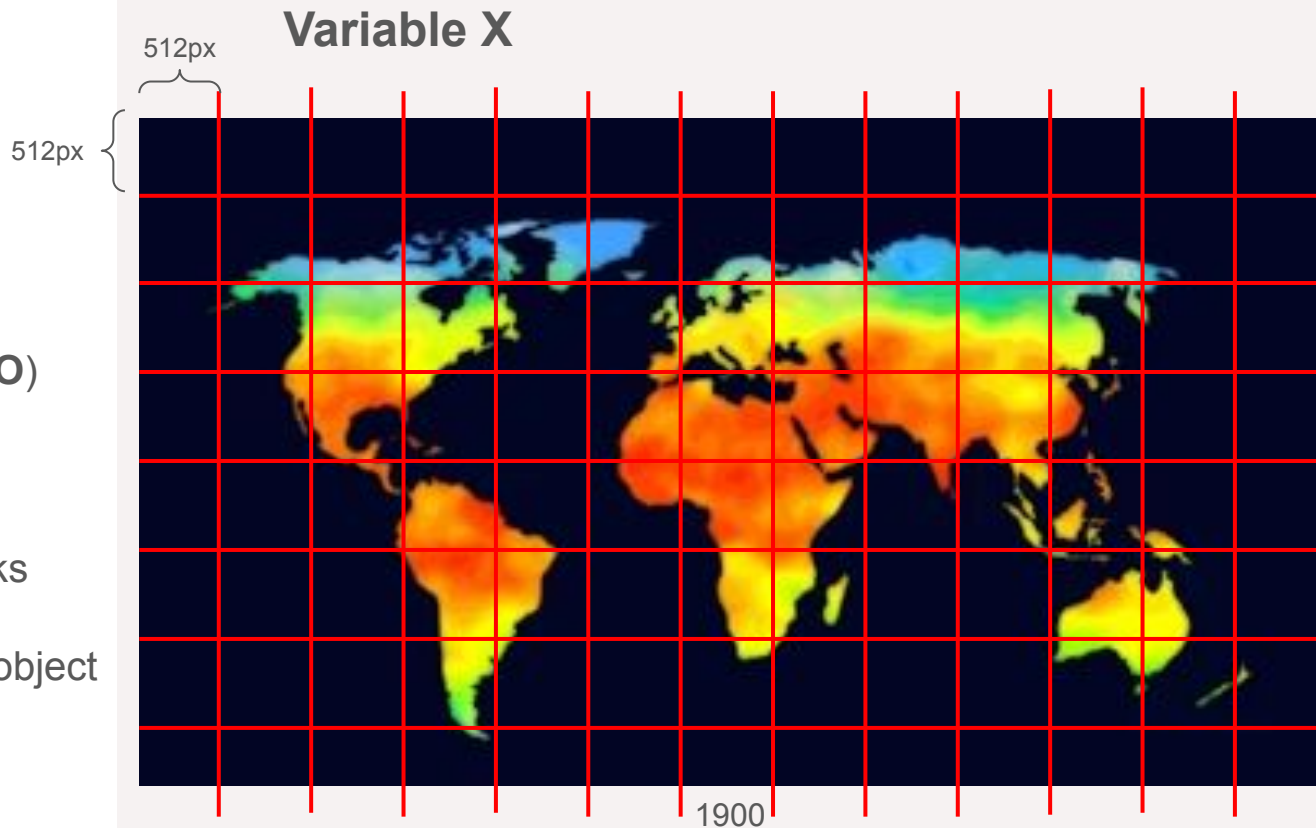


...



Zarr Stores

- Analysis-Ready
Cloud-Optimized (**ARCO**)
- Chunks data
- Stores > arrays > chunks
- Each store is a python object



Chunking → time: 1; lat: 512; long: 512

Zarr Stores

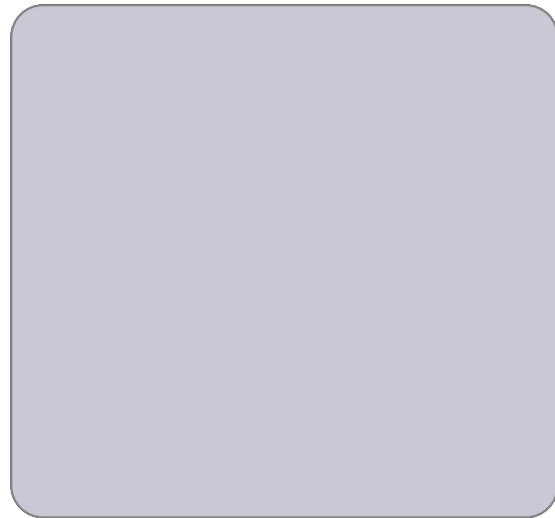
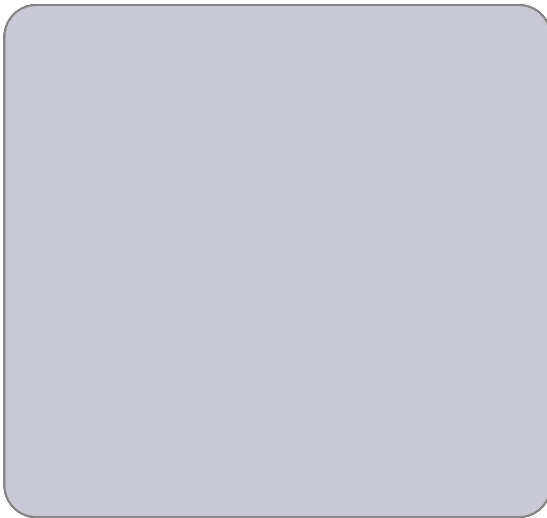
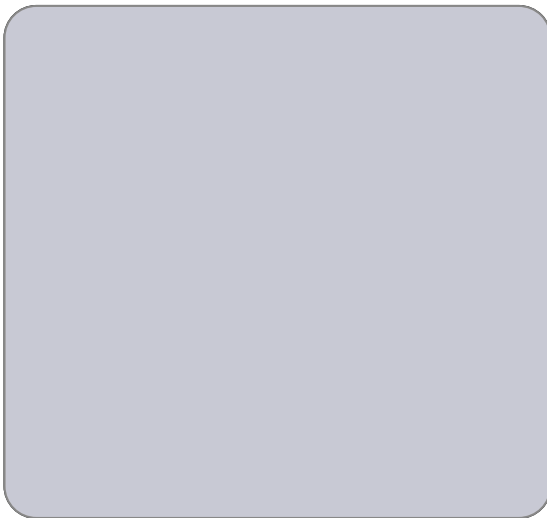
MultiZarr Stores

Chunking → time: 1; lat: 512;
lon: 512

Range → 1850 - 2020

Size → 660 GB

- LUH2
 - vars
- HILDA
 - vars
- Static
 - vars
- Koppen-Geiger
 - vars
- Population
 - vars



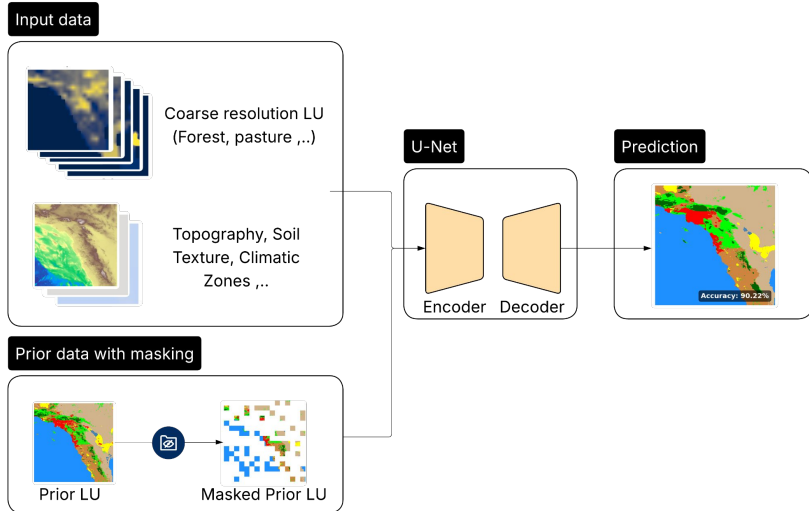
Zarr Stores

```
bsc096444 .../ai4land/CONCERTO/luh2_1850-2015_cube_512.zarr
(ins)> ll
total 458774
drwxr-sr-x 19 bsc406752 ehpc606 4096 Mar 3 10:09 ./
drwxr-sr-x 4 bsc406752 ehpc606 4096 Apr 28 09:33 ../
-rw-r--r-- 1 bsc406752 ehpc606 2 Mar 3 10:09 .zattrs
-rw-r--r-- 1 bsc406752 ehpc606 22 Mar 3 10:09 .zgroup
-rw-r--r-- 1 bsc406752 ehpc606 9284 Mar 3 10:09 .zmetadata
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:54 c3ann/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:55 c3nrx/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:54 c3per/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:54 c4ann/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:54 c4per/
drwxr-sr-x 2 bsc406752 ehpc606 4096 Mar 3 10:09 latitude/
drwxr-sr-x 2 bsc406752 ehpc606 4096 Mar 3 10:09 longitude/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:55 pastr/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:53 primf/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:53 primn/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:55 range/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:53 secdf/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:54 secdn/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:55 secma/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:55 secmb/
drwxr-sr-x 2 bsc406752 ehpc606 4096 Mar 3 10:09 time/
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:54 urban/
```

```
bsc096444 .../CONCERTO/luh2_1850-2015_cube_512.zarr/c3ann 09:12
(ins)> ll | head -50
total 56835026
drwxr-sr-x 2 bsc406752 ehpc606 16777216 Mar 3 17:54 ./
drwxr-sr-x 19 bsc406752 ehpc606 4096 Mar 3 10:09 ../
-rw-r--r-- 1 bsc406752 ehpc606 333 Mar 3 10:09 .zarray
-rw-r--r-- 1 bsc406752 ehpc606 76 Mar 3 10:09 .zattrs
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.0
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.1
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.10
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.11
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.12
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.13
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.14
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.15
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.16
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.17
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.18
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.19
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.2
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.20
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.21
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.22
-rw-r--r-- 1 bsc406752 ehpc606 4268 Mar 3 10:10 0.0.23
```

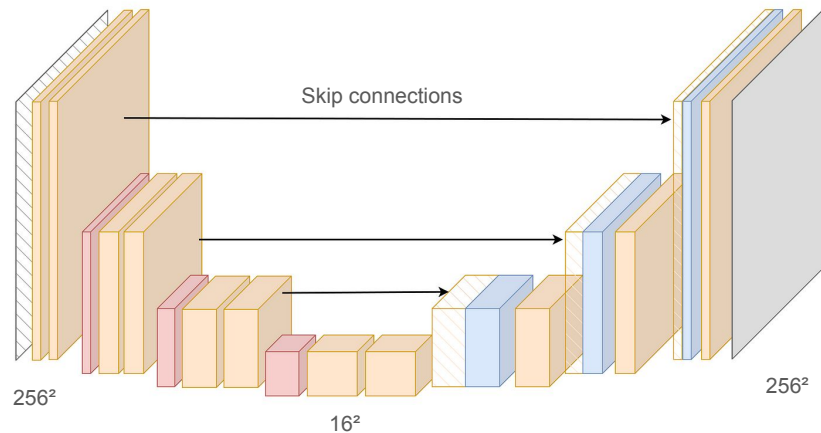
Training Pipeline

Pipeline



- 1 node = 4× H100
- GPFS underneath
- 12 workers/rank

U-Net



- 256x256 dimensions
- batch = 8 samples
- 9.5 MB per sample (uncompressed)

Reproducible Workflow

Goals

- Faster iteration (on code and dependencies)
- Reproducibility of experiments
- Easier understanding of the code (ruff)

	Storage	Reproducible	Env Creation Time
Conda	GB per env	No	Minutes
UV	Shared cache	Yes	Seconds

Initial Profiling

NSys



Zarr Stores

MultiZarr Stores

Chunking → time: 1; lat: 512;
lon: 512

Range → 1850 - 2020

Size → 660 GB

- LUH2
 - vars
- HILDA
 - vars
- Static
 - vars
- Koppen-Geiger
 - vars
- Population
 - vars

SingleZarr Store (Small)

Chunking → time: 1; lat: 512;
lon: 512

Range → 2000 - 2015

Size → 66 GB

- Data
 - LUH2 vars
 - HILDA vars
 - Static vars
 - KG vars
 - Population vars

SingleZarr Store (Big)

Chunking → time: 1; lat: 512;
lon: 512

Range → 1960 - 2015

Size → 213 GB

- Same as SingleZarr small

Zarr Stores

```
bsc096444 ~ 09:58
(ins)> ll /gpfs/scratch/ehpc736/data/AI4LAND_NON_preprocessed-data-2000-2015.zarr/
total 71206
drwxr-sr-x 28 bsc242971 ehpc736      4096 Apr 29 15:42 ./
drwxr-sr-x  9 bsc242971 ehpc736      4096 Jun 22 16:41 ../
-rw-r--r--  1 bsc242971 ehpc736         2 Apr 29 14:29 .zattrs
-rw-r--r--  1 bsc242971 ehpc736        24 Apr 29 14:29 .zgroup
-rw-r--r--  1 bsc242971 ehpc736    23299 Apr 29 14:29 .zmetadata
drwxr-sr-x  2 bsc242971 ehpc736   131072 Apr 29 14:49 ASPECT/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:42 LULC_states/
drwxr-sr-x  2 bsc242971 ehpc736   131072 Apr 29 14:57 SLOPE/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:29 c3ann/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:12 c3nfx/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:05 c3per/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 14:42 c4ann/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:09 c4per/
drwxr-sr-x  2 bsc242971 ehpc736   131072 Apr 29 14:48 elevation/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:45 kg_class/
drwxr-sr-x  2 bsc242971 ehpc736      4096 Apr 29 14:42 latitude/
drwxr-sr-x  2 bsc242971 ehpc736      4096 Apr 29 15:05 longitude/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 14:52 number_of_people/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:33 pastr/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:20 primf/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 14:47 primn/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 14:36 range/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:25 secdf/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:01 secdn/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:16 secma/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 15:37 secmb/
drwxr-sr-x  2 bsc242971 ehpc736      4096 Apr 29 14:57 time/
drwxr-sr-x  2 bsc242971 ehpc736   2097152 Apr 29 14:56 urban/
drwxr-sr-x  2 bsc242971 ehpc736   131072 Apr 29 14:42 weighted_ORGANIC/
drwxr-sr-x  2 bsc242971 ehpc736   131072 Apr 29 15:42 weighted_PCT_CLAY/
drwxr-sr-x  2 bsc242971 ehpc736   131072 Apr 29 14:57 weighted_PCT_SAND/
```

First Hypothesis

Single Store



- 80s spikes (multizarr) → 10s spikes (singlezarr)
- 660GB (multizarr) → 66GB (singlezarr)
- 10s spikes (single) → 25s spikes (big single)

Synchronizer

- Remove synchronizer object
- Measure time to retrieve a single sample:
 - synch on - 1177 ms
 - synch off - 1178 ms
- `sem_wait`'s stay
- Synchronizer is not used

Results

Isolating the Dataloader

Phase I

Sample time 1250 ms

- 14 LUH2 (x2) → 68%
- Population (x2) → 8.8%
- HILDA+ target → 3.9%
- HILDA+ prior → 3.8%
- Köppen-Geiger (x2) → 6.9%
- Static → 8.4 %



Phase II

Inside each store

- loading → 92%
- preprocessing → 8%



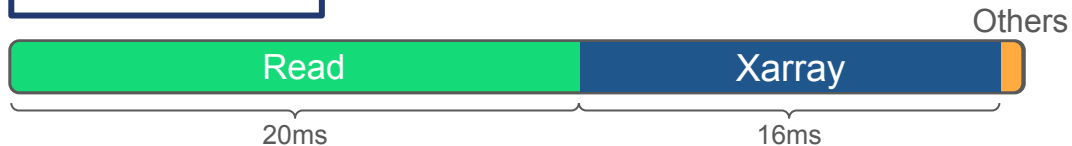
Phase III

Scalability by workers (w)

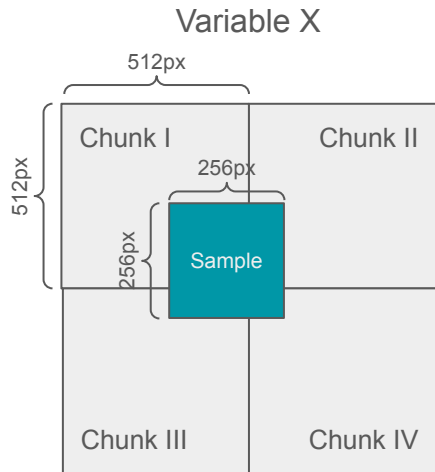
- 1 w → 0.9 samples / s
- 4 w → 3.3 samples / s
- 12 w → 9.6 samples / s

Diagnosis

Read of 1KB file



How much data we read?



- ~ **2.25** files per sample per variable
 - Total **93** files to read per sample
 - **3MB** (compressed)
- x200 times slower**

What can MN5 offer?

- **Reading bandwidth:** 500 MB/s
- **Cache miss trip (latency):**



- Invariant against store size
- We **miss 99%** of files

Zarr Stores

MultiZarr Stores

Chunking → time: 1; lat: 512;
lon: 512

Range → 1850 - 2020

Size → 660 GB

- LUH2
 - vars
- HILDA
 - vars
- Static
 - vars
- Koppen-Geiger
 - vars
- Population
 - vars

SingleZarr Store (Small)

Chunking → time: 1; lat: 512;
lon: 512

Range → 2000 - 2015

Size → 66 GB

- Data
 - LUH2 vars
 - HILDA vars
 - Static vars
 - KG vars
 - Population vars

SingleZarr Store (Big)

Chunking → time: 1; lat: 512;
lon: 512

Range → 1960 - 2015

Size → 213 GB

- Same as SingleZarr small

Merged Store

Chunking → per array

Range → 2000 - 2015

Size → 59 GB

- Data
 - dynamic chunks → variable: 17; time: 1; lat: 512; lon: 512
 - static chunks → variable: 6; lat: 512; lon: 512

Zarr Stores

```
bsc096444 ~ 10:04
(ins)> ll /gpfs/scratch/ehpc736/data/AI4LAND_merged-NON_preprocessed-data-2000-2015.zarr/
total 21
drwxr-sr-x 4 bsc242971 ehpc736 4096 Jun 11 12:54 ./
drwxr-sr-x 9 bsc242971 ehpc736 4096 Jun 22 16:41 ../
-rw-r--r-- 1 bsc242971 ehpc736 24 Jun 11 12:28 .zgroup
-rw-r--r-- 1 bsc242971 ehpc736 7148 Jun 11 12:54 .zmetadata
drwxr-sr-x 7 bsc242971 ehpc736 4096 Jun 11 12:52 dynamic/
drwxr-sr-x 6 bsc242971 ehpc736 4096 Jun 11 12:54 static/
```

```
bsc096444 ~ 10:06
(ins)> ll /gpfs/scratch/ehpc736/data/AI4LAND_merged-NON_preprocessed-data-2000-2015.zarr/dynamic/data/ | head -50
total 56739570
drwxr-sr-x 2 bsc242971 ehpc736 2097152 Jun 11 12:52 ./
drwxr-sr-x 7 bsc242971 ehpc736 4096 Jun 11 12:52 ../
-rw-r--r-- 1 bsc242971 ehpc736 394 Jun 11 12:28 .zarray
-rw-r--r-- 1 bsc242971 ehpc736 112 Jun 11 12:28 .zattrs
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.0
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.1
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.10
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.11
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.12
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.13
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.14
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.15
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.16
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.17
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.18
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.19
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.2
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.20
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.21
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.22
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.23
-rw-r--r-- 1 bsc242971 ehpc736 71960 Jun 11 12:50 0.0.0.24
```

Solution:

Merged Store

- 14 LUH2
 - Population
 - KG
 - HILDA
- } Dynamic

9 files / sample

- Aspect
 - Elevation
 - Slope
 - 3 Soil
- } Static

Isolated Dataloader

Sample time: 1250 ms → 170 ms (x7)

	SingleZarr (samples / s)	MergedZarr (samples / s)
1 worker	0.8	5.9
4 workers	3.3	25.3
12 workers	9.6	70.5

x7

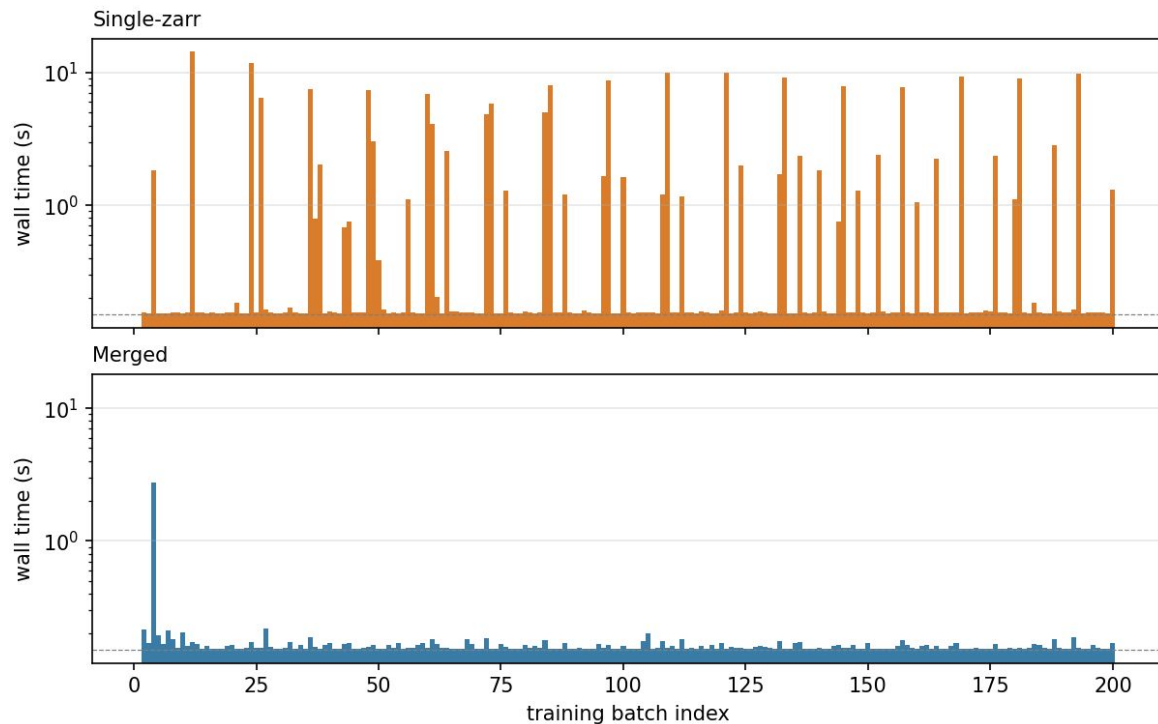
GPU requires 53 samples / s !!!

Profiling



Solution: Merged Store

Training



Total: 354 s

x4.6

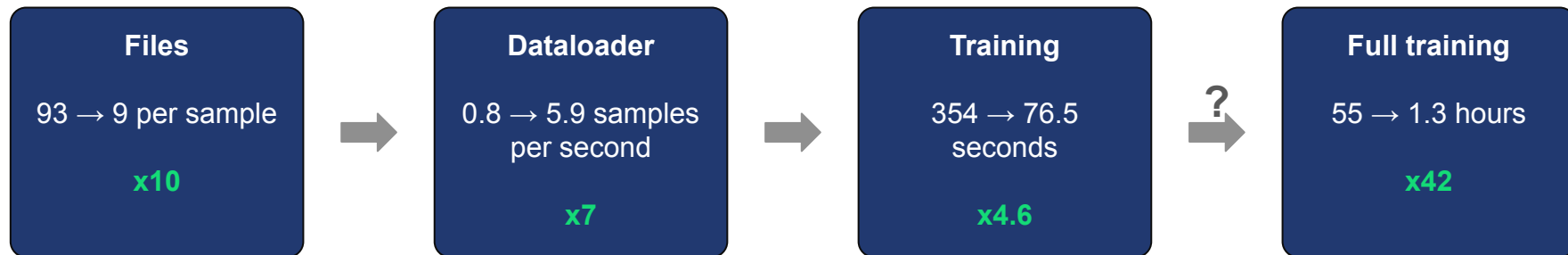
Total: 76.5 s

Solution: Merged Store

Scalability

		Single-zarr			Merged				
Ranks	Workers	Samp/s	Wall (s)	Start (s)	Samp/s	Wall (s)	Start (s)		
Weak Scaling	1	Strong Scaling { 8 12 20 }	100% 86%	6	2447	58	50	397	104
				9	1774	96	51	392	98
				13	1327	95	52	408	105
	4	8	19	2976	121	167	486	120	
		12	28	2106	69	196	434	115	
		20	37	1785	80	192	454	114	
	8	8	38	3038	112	367 [†]	453	120	
		12	53	2213	82	364 [†]	436	100	
		20	71	1849	105	358	454	82	

Solution: Summary



Projection

Assumptions

- we create merged store from 1900 - 2020
- merged store can deliver 53 samples / s at full window

Full training → 10 epochs
3000 batches per epoch
30500 total batches (500 test)

Single Store

- 1 in 12 is long
- 2540 long batches
- ~75s per long batch
- 27960 short batches at ~0.15s each

55 hours

Merged Store

- No stalling
- ~0.15s per batch

1.3 hours

Conclusions

MAIN

Achieve a faster training time

Achieved x4.6
(potentially x40)



Secondary

Faster training,
same accuracy

Same data
Same loss



Understand each
speedup

Latency is
main issue



Reinvest the
savings

Train more
samples



Reusable
guidelines

Careful
chunking



The End (?)

Future Work

- Full-window merged store: measure speedup + reinvesting the savings metrics improvement
- Find store size effect
- Data: bigger chunks (1024^2), fold years along time, raw Zarr path, Zarr v3 async
- Compute side: learn from Anemoi (the real next limit)
- Paper on interoperability vs performance of data



Thank you all!

Earth Sciences
Department



**Barcelona
Supercomputing
Center**
Centro Nacional de Supercomputación

Zarr vs NetCDF

	NetCDF-4 / HDF5	Zarr
<i>Storage model</i>	One monolithic file	One object per chunk + JSON metadata
<i>Parallel reads</i>	Serialized through one file/lock	Independent per chunk
<i>Random / partial reads</i>	Open and seek the whole container	Fetch only the covering chunks
<i>Parallel writes</i>	MPI-IO; blocked by compression	Lock-free, one key per writer
<i>Cloud / object storage</i>	Poor (no cheap seek inside a file)	Native (chunk = object)
<i>Standards & portability</i>	Strong: CF conventions, single file	Younger; many files per store
<i>Many small files on HPC</i>	Not an issue	Metadata-server pressure